T3 group factors: UV

Active theory: SM + S_1 + S_2 + F. The same beta function is shown across runs in a single comparison table, with exact analytic group factors in the run columns.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Field representations

run	S_1	S_2	F	α
T3 − A − m4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	-4

$\mathrm{T3-A-m2}$	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	-2
$\mathrm{T3-A-p0}$	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	0
$\mathrm{T3-B-m3}$	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	-3
$\mathrm{T3-B-m1}$	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	-1
$\mathrm{T3-B-p1}$	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	1
$\mathrm{T3-C-m3}$	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	-3
$\mathrm{T3-C-m1}$	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	-1
$\mathrm{T3-C-p1}$	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	1
$\mathrm{T3-D-m2}$	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	-2
$\mathrm{T3-D-p0}$	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	0
$\mathrm{T3-D-p2}$	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	2
$\mathrm{T3-E-m4}$	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	-4
$\mathrm{T3-E-m2}$	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	-2
$\mathrm{T3-E-p0}$	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	0
$\mathrm{T3-E-p2}$	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	2

Gauge and Yukawa beta functions

$\beta^{(1)}_{g_1}$

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-}$
g_1^3	$\frac{163}{6}$	$\frac{55}{6}$	$\frac{59}{6}$	

$\mathcal{S}$	$\mathrm{T3-B-p1}\ (\alpha=1)$	$\mathrm{T3-C-m3}\ (\alpha=-3)$	$\mathrm{T3-C-m1}\ (\alpha=-1)$	$\mathrm{T3-C-}$
g_1^3	$\frac{25}{2}$	$\frac{41}{2}$	$\frac{43}{6}$	$\frac{4}{2}$

$\mathcal{S}$	$\mathrm{T3-D-p0}\ (\alpha=0)$	$\mathrm{T3-D-p2}\ (\alpha=2)$	$\mathrm{T3-E-m4}\ (\alpha=-4)$	$\mathrm{T3-E-m2}$
g_1^3	$\frac{55}{6}$	$\frac{163}{6}$	$\frac{179}{6}$	$\frac{59}{6}$

$\mathcal{S}$	$\mathrm{T3-E-p2}\ (\alpha=2)$
g_1^3	$\frac{179}{6}$

$\beta^{(1)}_{g_2}$

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-}$
g_2^3	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	

$\mathcal{S}$	$\mathrm{T3-B-p1}\ (\alpha=1)$	$\mathrm{T3-C-m3}\ (\alpha=-3)$	$\mathrm{T3-C-m1}\ (\alpha=-1)$	$\mathrm{T3-C-}$
g_2^3	$-\frac{17}{6}$	$\frac{31}{6}$	$\frac{7}{6}$	$\frac{3}{6}$

$\mathcal{S}$	$\mathrm{T3-D-p0}\ (\alpha=0)$	$\mathrm{T3-D-p2}\ (\alpha=2)$	$\mathrm{T3-E-m4}\ (\alpha=-4)$	$\mathrm{T3-E-m2}$
g_2^3	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{6}$

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
g_2^3	$\frac{1}{6}$

$\beta_{g_3}^{(1)}$

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B −
g_3^3	-7	-7	-7	

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C −
g_3^3	-7	-7	-7	$-$

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2
g_3^3	-7	-7	-7	-7

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
g_3^3	-7

$\beta_{Y_u}^{(1)}$

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B −
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	
$g_3^2(Y_u)_{ij}$	-8	-8	-8	
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-$
$T(Y_u)_{ij}$	1	1	1	

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C −
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	$-$
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-$
$T(Y_u)_{ij}$	1	1	1	1

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1

$\beta^{(1)}_{Y_d}$

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B −
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	
$g_3^2(Y_d)_{ij}$	-8	-8	-8	
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	
$T(Y_d)_{ij}$	1	1	1	

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C −
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	
$g_3^2(Y_d)_{ij}$	-8	-8	-8	
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	
$T(Y_d)_{ij}$	1	1	1	

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1	1

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

$\beta^{(1)}_{Y_e}$

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B −
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	

$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C −
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-$
$T(Y_e)_{ij}$	1	1	1	1

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{3}{4}$
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1

$\beta_{y_1}^{(1)}$

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B −
$T_\nu^{(1)}(y_1)_{ij}$	2	2	2	
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	
$(y_1(y_1)^\dagger y_1)_{ij}$	1	1	1	
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	
$g_1^2(y_1)_{ij}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-$
$\mathrm{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	0	0	
$(y_1(y_2)^\dagger y_2)_{ij}$	0	0	0	

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C −
$T_\nu^{(1)}(y_1)_{ij}$	1	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$	$\frac{5}{4}$	$\frac{5}{4}$	$\frac{5}{4}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{4}$	$-\frac{33}{4}$	$-\frac{33}{4}$	$-$
$g_1^2(y_1)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{3}{4}$	$-$

$\mathrm{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	0	$\frac{3}{2}$	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0	0	$\frac{1}{2}$	0

$\mathcal{S}$	$\mathrm{T3-D-p0}\ (\alpha=0)$	$\mathrm{T3-D-p2}\ (\alpha=2)$	$\mathrm{T3-E-m4}\ (\alpha=-4)$	$\mathrm{T3-E-m2}$
$T_\nu^{(1)}(y_1)_{ij}$	1	1	1	1
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(y_1)_{ij}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\mathrm{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	0	0	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0	0	0	0

$\mathcal{S}$	$\mathrm{T3-E-p2}\ (\alpha=2)$
$T_\nu^{(1)}(y_1)_{ij}$	1
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$
$g_1^2(y_1)_{ij}$	$-\frac{15}{2}$
$\mathrm{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0

$\beta_{y_2}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-}$
$T_\nu^{(2)}(y_2)_{ij}$	1	1	1	
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	
$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	
$g_1^2(y_2)_{ij}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	0	0	
$(y_2(y_1)^\dagger y_1)_{ij}$	0	0	0	

$\mathcal{S}$	$\mathrm{T3-B-p1}\ (\alpha=1)$	$\mathrm{T3-C-m3}\ (\alpha=-3)$	$\mathrm{T3-C-m1}\ (\alpha=-1)$	$\mathrm{T3-C-}$
$T_\nu^{(2)}(y_2)_{ij}$	1	$\frac{3}{2}$	$\frac{3}{2}$	
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$	$\frac{5}{4}$	$\frac{5}{4}$	
$g_2^2(y_2)_{ij}$	$-\frac{9}{4}$	$-\frac{33}{4}$	$-\frac{33}{4}$	
$g_1^2(y_2)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{3}{4}$	
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	0	$\frac{3}{2}$	
$(y_2(y_1)^\dagger y_1)_{ij}$	0	0	$\frac{1}{2}$	

$\mathcal{S}$	$\mathrm{T3-D-p0}\ (\alpha=0)$	$\mathrm{T3-D-p2}\ (\alpha=2)$	$\mathrm{T3-E-m4}\ (\alpha=-4)$	$\mathrm{T3-E-m2}$
$T_\nu^{(2)}(y_2)_{ij}$	2	2	1	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$

$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	1	1	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(y_2)_{ij}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	0	0	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0	0	0	0

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$T_\nu^{(2)}(y_2)_{ij}$	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$
$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$
$g_1^2(y_2)_{ij}$	$-\frac{15}{2}$
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0

β_{M_F}

Run block 1

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B − m3 ($\alpha =$
$(y_1^T y_1^*)M_F$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1
$M_F(y_2^\dagger y_2)$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	1
$g_2^2 M_F$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	0
$g_1^2 M_F$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	−6

Run block 2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
$(y_1^T y_1^*)M_F$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$M_F(y_2^\dagger y_2)$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$g_2^2 M_F$	0	−12	−12	−12
$g_1^2 M_F$	−6	−6	0	−6

Run block 3

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$(y_1^T y_1^*)M_F$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$M_F(y_2^\dagger y_2)$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$
$g_2^2 M_F$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 M_F$	$-\frac{3}{2}$	$-\frac{27}{2}$	$-\frac{27}{2}$	$-\frac{3}{2}$

Run block 4

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$(y_1^T y_1^*)M_F$	$\frac{3}{4}$
$M_F(y_2^\dagger y_2)$	$\frac{3}{4}$
$g_2^2 M_F$	$-\frac{9}{2}$
$g_1^2 M_F$	$-\frac{27}{2}$

$\beta_{m_1^2}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha=-4$)	T3 − A − m2 ($\alpha=-2$)	T3 − A − p0 ($\alpha=0$)	T3 − B − m3 ($\alpha=$
$m_1^2\,T_\nu^{(1)}$	4	4	4	2
$M_F^\dagger M_F\,T_\nu^{(1)}$	−8	−8	−8	−4
$\lambda_{S_1}^{(1)}\,m_1^2$	4	4	4	6
$g_2^2\,m_1^2$	0	0	0	$-\frac{9}{2}$
$g_1^2\,m_1^2$	−24	−6	0	$-\frac{27}{2}$
$\lambda_{12}^{(1)}\,m_2^2$	6	6	6	4
$\lambda_{S_1}^{(A)}\,m_1^2$	0	0	0	0
$\lambda_{12}^{(\times)}\,m_2^2$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha=1$)	T3 − C − m3 ($\alpha=-3$)	T3 − C − m1 ($\alpha=-1$)	T3 − C − p1 ($\alpha=$
$m_1^2\,T_\nu^{(1)}$	2	3	3	3
$M_F^\dagger M_F\,T_\nu^{(1)}$	−4	−6	−24	−6
$\lambda_{S_1}^{(1)}\,m_1^2$	6	6	6	6
$g_2^2\,m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2\,m_1^2$	$-\frac{3}{2}$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)}\,m_2^2$	4	4	4	4
$\lambda_{S_1}^{(A)}\,m_1^2$	0	0	0	0
$\lambda_{12}^{(\times)}\,m_2^2$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha=0$)	T3 − D − p2 ($\alpha=2$)	T3 − E − m4 ($\alpha=-4$)	T3 − E − m2 ($\alpha=-$
$m_1^2\,T_\nu^{(1)}$	2	2	2	2
$M_F^\dagger M_F\,T_\nu^{(1)}$	−4	−4	−4	−4
$\lambda_{S_1}^{(1)}\,m_1^2$	8	8	8	8
$g_2^2\,m_1^2$	−12	−12	−12	−12
$g_1^2\,m_1^2$	0	−6	−24	−6
$\lambda_{12}^{(1)}\,m_2^2$	2	2	6	6
$\lambda_{S_1}^{(A)}\,m_1^2$	4	4	4	4
$\lambda_{12}^{(\times)}\,m_2^2$	0	0	4	4

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha=2$)
$m_1^2\,T_\nu^{(1)}$	2
$M_F^\dagger M_F\,T_\nu^{(1)}$	−4
$\lambda_{S_1}^{(1)}\,m_1^2$	8
$g_2^2\,m_1^2$	−12
$g_1^2\,m_1^2$	−6
$\lambda_{12}^{(1)}\,m_2^2$	6
$\lambda_{S_1}^{(A)}\,m_1^2$	4
$\lambda_{12}^{(\times)}\,m_2^2$	4

$\beta_{m_2^2}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-m3}\ (\alpha=$
$m_2^2\,T_\nu^{(2)}$	2	2	2	2
$M_F^\dagger M_F\,T_\nu^{(2)}$	-4	-4	-4	-4
$\lambda_{S_2}^{(1)}\,m_2^2$	8	8	8	6
$g_2^2\,m_2^2$	-12	-12	-12	$-\frac{9}{2}$
$g_1^2\,m_2^2$	-6	0	-6	$-\frac{3}{2}$
$\lambda_{12}^{(1)}\,m_1^2$	2	2	2	4
$\lambda_{S_2}^{(A)}\,m_2^2$	4	4	4	0
$\lambda_{12}^{(\times)}\,m_1^2$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	$\mathrm{T3-B-p1}\ (\alpha=1)$	$\mathrm{T3-C-m3}\ (\alpha=-3)$	$\mathrm{T3-C-m1}\ (\alpha=-1)$	$\mathrm{T3-C-p1}\ (\alpha=$
$m_2^2\,T_\nu^{(2)}$	2	3	3	3
$M_F^\dagger M_F\,T_\nu^{(2)}$	-4	-6	-24	-6
$\lambda_{S_2}^{(1)}\,m_2^2$	6	6	6	6
$g_2^2\,m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2\,m_2^2$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{27}{2}$
$\lambda_{12}^{(1)}\,m_1^2$	4	4	4	4
$\lambda_{S_2}^{(A)}\,m_2^2$	0	0	0	0
$\lambda_{12}^{(\times)}\,m_1^2$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	$\mathrm{T3-D-p0}\ (\alpha=0)$	$\mathrm{T3-D-p2}\ (\alpha=2)$	$\mathrm{T3-E-m4}\ (\alpha=-4)$	$\mathrm{T3-E-m2}\ (\alpha=-$
$m_2^2\,T_\nu^{(2)}$	4	4	2	2
$M_F^\dagger M_F\,T_\nu^{(2)}$	-8	-8	-4	-4
$\lambda_{S_2}^{(1)}\,m_2^2$	4	4	8	8
$g_2^2\,m_2^2$	0	0	-12	-12
$g_1^2\,m_2^2$	-6	-24	-6	0
$\lambda_{12}^{(1)}\,m_1^2$	6	6	6	6
$\lambda_{S_2}^{(A)}\,m_2^2$	0	0	4	4
$\lambda_{12}^{(\times)}\,m_1^2$	0	0	4	4

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	$\mathrm{T3-E-p2}\ (\alpha=2)$
$m_2^2\,T_\nu^{(2)}$	2
$M_F^\dagger M_F\,T_\nu^{(2)}$	-4
$\lambda_{S_2}^{(1)}\,m_2^2$	8
$g_2^2\,m_2^2$	-12
$g_1^2\,m_2^2$	-24
$\lambda_{12}^{(1)}\,m_1^2$	6
$\lambda_{S_2}^{(A)}\,m_2^2$	4
$\lambda_{12}^{(\times)}\,m_1^2$	4

$\beta_{\lambda_{HS_1}^{(1)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha=-4$)	T3 − A − m2 ($\alpha=-2$)	T3 − A − p0 ($\alpha=0$)	T3 − B − m3 ($\alpha=$
$\lambda^{(1)}_{HS_1}\, T^{(1)}_\nu$	4	4	4	2
$\mathrm{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	-4	-4	-4	-2
$\lvert\lambda_5\rvert^2$	12	12	12	6
$g_2^2\,\lambda^{(1)}_{HS_1}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	-9
$g_1^2\,\lambda^{(1)}_{HS_1}$	$-\frac{51}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$	-15
$\lambda_1\,\lambda^{(1)}_{HS_1}$	6	6	6	6
$\left(\lambda^{(1)}_{HS_1}\right)^2$	4	4	4	4
$\lambda^{(1)}_{HS_1}\,\lambda^{(1)}_{S_1}$	4	4	4	6
$\lambda^{(1)}_{12}\,\lambda^{(1)}_{HS_2}$	6	6	6	4
g_2^4	0	0	0	$\frac{9}{4}$
g_1^4	12	3	0	$\frac{27}{4}$
$\lambda^{(1)}_{HS_1}\, T$	2	2	2	2
$\left(\lambda^{(A)}_{HS_1}\right)^2$	0	0	0	$\frac{3}{4}$
$\lambda^{(1)}_{HS_1}\,\lambda^{(A)}_{S_1}$	0	0	0	0
$\lambda^{(1)}_{HS_2}\,\lambda^{(\times)}_{12}$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha=1$)	T3 − C − m3 ($\alpha=-3$)	T3 − C − m1 ($\alpha=-1$)	T3 − C − p1 ($\alpha=$
$\lambda^{(1)}_{HS_1}\, T^{(1)}_\nu$	2	3	3	3
$\mathrm{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	-2	-3	-3	-3
$\lvert\lambda_5\rvert^2$	6	6	6	6
$g_2^2\,\lambda^{(1)}_{HS_1}$	-9	-9	-9	-9
$g_1^2\,\lambda^{(1)}_{HS_1}$	-3	-15	-3	-3
$\lambda_1\,\lambda^{(1)}_{HS_1}$	6	6	6	6
$\left(\lambda^{(1)}_{HS_1}\right)^2$	4	4	4	4
$\lambda^{(1)}_{HS_1}\,\lambda^{(1)}_{S_1}$	6	6	6	6
$\lambda^{(1)}_{12}\,\lambda^{(1)}_{HS_2}$	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	$\frac{3}{4}$	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda^{(1)}_{HS_1}\, T$	2	2	2	2
$\left(\lambda^{(A)}_{HS_1}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda^{(1)}_{HS_1}\,\lambda^{(A)}_{S_1}$	0	0	0	0
$\lambda^{(1)}_{HS_2}\,\lambda^{(\times)}_{12}$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha=0$)	T3 − D − p2 ($\alpha=2$)	T3 − E − m4 ($\alpha=-4$)	T3 − E − m2 ($\alpha=-$
$\lambda^{(1)}_{HS_1}\, T^{(1)}_\nu$	2	2	2	2
$\mathrm{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	-2	-2	-2	-2
$\lvert\lambda_5\rvert^2$	4	4	4	4
$g_2^2\,\lambda^{(1)}_{HS_1}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\,\lambda^{(1)}_{HS_1}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{15}{2}$
$\lambda_1\,\lambda^{(1)}_{HS_1}$	6	6	6	6
$\left(\lambda^{(1)}_{HS_1}\right)^2$	4	4	4	4
$\lambda^{(1)}_{HS_1}\,\lambda^{(1)}_{S_1}$	8	8	8	8
$\lambda^{(1)}_{12}\,\lambda^{(1)}_{HS_2}$	2	2	6	6
g_2^4	6	6	6	6
g_1^4	0	3	12	3

$\lambda^{(1)}_{HS_1}\, T$	2	2	2	2
$\left(\lambda^{(A)}_{HS_1}\right)^2$	2	2	2	2
$\lambda^{(1)}_{HS_1}\, \lambda^{(A)}_{S_1}$	4	4	4	4
$\lambda^{(1)}_{HS_2}\, \lambda^{(\times)}_{12}$	0	0	4	4

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	$\mathrm{T3-E-p2}\ (\alpha=2)$
$\lambda^{(1)}_{HS_1}\, T^{(1)}_{\nu}$	2
$\mathrm{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	-2
$\lvert\lambda_5\rvert^2$	4
$g_2^2\, \lambda^{(1)}_{HS_1}$	$-\frac{33}{2}$
$g_1^2\, \lambda^{(1)}_{HS_1}$	$-\frac{15}{2}$
$\lambda_1\, \lambda^{(1)}_{HS_1}$	6
$\left(\lambda^{(1)}_{HS_1}\right)^2$	4
$\lambda^{(1)}_{HS_1}\, \lambda^{(1)}_{S_1}$	8
$\lambda^{(1)}_{12}\, \lambda^{(1)}_{HS_2}$	6
g_2^4	6
g_1^4	3
$\lambda^{(1)}_{HS_1}\, T$	2
$\left(\lambda^{(A)}_{HS_1}\right)^2$	2
$\lambda^{(1)}_{HS_1}\, \lambda^{(A)}_{S_1}$	4
$\lambda^{(1)}_{HS_2}\, \lambda^{(\times)}_{12}$	4

$\beta_{\lambda^{(1)}_{HS_2}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-m3}\ (\alpha=$
$\lambda^{(1)}_{HS_2}\, T^{(2)}_{\nu}$	2	2	2	2
$\mathrm{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	-2	-2	-2	-2
$\lvert\lambda_5\rvert^2$	4	4	4	6
$g_2^2\, \lambda^{(1)}_{HS_2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	-9
$g_1^2\, \lambda^{(1)}_{HS_2}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$	-3
$\lambda_1\, \lambda^{(1)}_{HS_2}$	6	6	6	6
$\left(\lambda^{(1)}_{HS_2}\right)^2$	4	4	4	4
$\lambda^{(1)}_{HS_2}\, \lambda^{(1)}_{S_2}$	8	8	8	6
$\lambda^{(1)}_{12}\, \lambda^{(1)}_{HS_1}$	2	2	2	4
g_2^4	6	6	6	$\frac{9}{4}$
g_1^4	3	0	3	$\frac{3}{4}$
$\left(\lambda^{(A)}_{HS_2}\right)^2$	2	2	2	$\frac{3}{4}$
$\lambda^{(1)}_{HS_2}\, \lambda^{(A)}_{S_2}$	4	4	4	0
$\lambda^{(1)}_{HS_2}\, T$	2	2	2	2
$\lambda^{(1)}_{HS_1}\, \lambda^{(\times)}_{12}$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	$\mathrm{T3-B-p1}\ (\alpha=1)$	$\mathrm{T3-C-m3}\ (\alpha=-3)$	$\mathrm{T3-C-m1}\ (\alpha=-1)$	$\mathrm{T3-C-p1}\ (\alpha=$
$\lambda^{(1)}_{HS_2}\, T^{(2)}_{\nu}$	2	3	3	3
$\mathrm{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	-2	-3	-3	-3

$\lvert\lambda_5\rvert^2$	6	6	6	6
$g_2^2\,\lambda_{HS_2}^{(1)}$	-9	-9	-9	-9
$g_1^2\,\lambda_{HS_2}^{(1)}$	-15	-3	-3	-15
$\lambda_1\,\lambda_{HS_2}^{(1)}$	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(1)}$	6	6	6	6
$\lambda_{12}^{(1)}\,\lambda_{HS_1}^{(1)}$	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
$\left(\lambda_{HS_2}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(A)}$	0	0	0	0
$\lambda_{HS_2}^{(1)}\,T$	2	2	2	2
$\lambda_{HS_1}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha=0$)	T3 − D − p2 ($\alpha=2$)	T3 − E − m4 ($\alpha=-4$)	T3 − E − m2 ($\alpha=-$
$\lambda_{HS_2}^{(1)}\,T_\nu^{(2)}$	4	4	2	2
$\mathrm{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	-4	-4	-2	-2
$\lvert\lambda_5\rvert^2$	12	12	4	4
$g_2^2\,\lambda_{HS_2}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\,\lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\lambda_1\,\lambda_{HS_2}^{(1)}$	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(1)}$	4	4	8	8
$\lambda_{12}^{(1)}\,\lambda_{HS_1}^{(1)}$	6	6	6	6
g_2^4	0	0	6	6
g_1^4	3	12	3	0
$\left(\lambda_{HS_2}^{(A)}\right)^2$	0	0	2	2
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(A)}$	0	0	4	4
$\lambda_{HS_2}^{(1)}\,T$	2	2	2	2
$\lambda_{HS_1}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	4	4

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha=2$)
$\lambda_{HS_2}^{(1)}\,T_\nu^{(2)}$	2
$\mathrm{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	-2
$\lvert\lambda_5\rvert^2$	4
$g_2^2\,\lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$
$g_1^2\,\lambda_{HS_2}^{(1)}$	$-\frac{51}{2}$
$\lambda_1\,\lambda_{HS_2}^{(1)}$	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(1)}$	8
$\lambda_{12}^{(1)}\,\lambda_{HS_1}^{(1)}$	6
g_2^4	6
g_1^4	12
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2
$\lambda_{HS_2}^{(1)}\,\lambda_{S_2}^{(A)}$	4
$\lambda_{HS_2}^{(1)}\,T$	2

$\lambda_{HS_1}^{(1)}\,\lambda_{12}^{(\times)}$	4

$\beta_{\lambda_{12}^{(1)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha=-4$)	T3 − A − m2 ($\alpha=-2$)	T3 − A − p0 ($\alpha=0$)	T3 − B − m3 ($\alpha=$
$\lambda_{12}^{(1)}\,T_\nu^{(1)}$	4	4	4	0
$\lambda_{12}^{(1)}\,T_\nu^{(2)}$	2	2	2	0
$\mathrm{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	−4	−4	−4	−2
$\lvert\lambda_5\rvert^2$	4	4	4	3
$g_2^2\,\lambda_{12}^{(1)}$	−12	−12	−12	−9
$g_1^2\,\lambda_{12}^{(1)}$	−30	−6	−6	−15
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(1)}$	4	4	4	6
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(1)}$	8	8	8	6
$\lambda_{HS_1}^{(1)}\,\lambda_{HS_2}^{(1)}$	4	4	4	4
g_2^4	0	0	0	$\frac{9}{4}$
g_1^4	48	0	0	$\frac{27}{4}$
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(A)}$	4	4	4	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	$\frac{3}{4}$
$\lambda_{12}^{(1)}\,T_\nu$	0	0	0	2
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha=1$)	T3 − C − m3 ($\alpha=-3$)	T3 − C − m1 ($\alpha=-1$)	T3 − C − p1 ($\alpha=$
$\lambda_{12}^{(1)}\,T_\nu^{(1)}$	0	0	0	0
$\lambda_{12}^{(1)}\,T_\nu^{(2)}$	0	0	0	0
$\mathrm{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	−2	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\lvert\lambda_5\rvert^2$	3	3	3	3
$g_2^2\,\lambda_{12}^{(1)}$	−9	−9	−9	−9
$g_1^2\,\lambda_{12}^{(1)}$	−15	−15	−3	−15
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(1)}$	6	6	6	6
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(1)}$	6	6	6	6
$\lambda_{HS_1}^{(1)}\,\lambda_{HS_2}^{(1)}$	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	$\frac{27}{4}$	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(A)}$	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{12}^{(1)}\,T_\nu$	2	3	3	3
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$\lambda_{12}^{(1)}\, T_\nu^{(1)}$	2	2	0	0
$\lambda_{12}^{(1)}\, T_\nu^{(2)}$	4	4	0	0
$\mathrm{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	-4	-4	-2	-2
$\lvert\lambda_5\rvert^2$	4	4	$\frac{4}{3}$	$\frac{4}{3}$
$g_2^2\, \lambda_{12}^{(1)}$	-12	-12	-24	-24
$g_1^2\, \lambda_{12}^{(1)}$	-6	-30	-30	-6
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4
$\lambda_{12}^{(1)}\, \lambda_{S_1}^{(1)}$	8	8	8	8
$\lambda_{12}^{(1)}\, \lambda_{S_2}^{(1)}$	4	4	8	8
$\lambda_{HS_1}^{(1)}\, \lambda_{HS_2}^{(1)}$	4	4	4	4
g_2^4	0	0	12	12
g_1^4	0	48	48	0
$\lambda_{12}^{(1)}\, \lambda_{S_2}^{(A)}$	0	0	4	4
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	4	4
$\lambda_{12}^{(1)}\, T_\nu$	0	0	2	2
$\lambda_{12}^{(1)}\, \lambda_{S_1}^{(A)}$	4	4	4	4
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	4	4
$\lambda_{12}^{(\times)}\, \lambda_{S_1}^{(A)}$	0	0	4	4
$\lambda_{12}^{(\times)}\, \lambda_{S_2}^{(A)}$	0	0	4	4

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$\lambda_{12}^{(1)}\, T_\nu^{(1)}$	0
$\lambda_{12}^{(1)}\, T_\nu^{(2)}$	0
$\mathrm{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	-2
$\lvert\lambda_5\rvert^2$	$\frac{4}{3}$
$g_2^2\, \lambda_{12}^{(1)}$	-24
$g_1^2\, \lambda_{12}^{(1)}$	-30
$\left(\lambda_{12}^{(1)}\right)^2$	4
$\lambda_{12}^{(1)}\, \lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)}\, \lambda_{S_2}^{(1)}$	8
$\lambda_{HS_1}^{(1)}\, \lambda_{HS_2}^{(1)}$	4
g_2^4	12
g_1^4	48
$\lambda_{12}^{(1)}\, \lambda_{S_2}^{(A)}$	4
$\left(\lambda_{12}^{(A)}\right)^2$	4
$\lambda_{12}^{(1)}\, T_\nu$	2
$\lambda_{12}^{(1)}\, \lambda_{S_1}^{(A)}$	4
$\left(\lambda_{12}^{(\times)}\right)^2$	4
$\lambda_{12}^{(\times)}\, \lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(\times)}\, \lambda_{S_2}^{(A)}$	4

β_{λ_5}

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B − m3 ($\alpha =$

$\lambda_5\,\lambda_1$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{HS_2}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{12}^{(1)}$	2	2	2	2
$g_2^2\,\lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$	-9
$g_1^2\,\lambda_5$	$-\frac{33}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	-9
$\lambda_5\,T_\nu^{(1)}$	2	2	2	0
$\lambda_5\,T_\nu^{(2)}$	1	1	1	0
$\lambda_5\,\lambda_{HS_2}^{(A)}$	4	4	4	2
$\lambda_5\,T$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(A)}$	0	0	0	-2
$\lambda_5\,\lambda_{12}^{(A)}$	0	0	0	$-\frac{1}{2}$
$\lambda_5\,T_\nu$	0	0	0	1
$\lambda_5\,\lambda_{12}^{(\times)}$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
$\lambda_5\,\lambda_1$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{HS_2}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{12}^{(1)}$	2	2	2	2
$g_2^2\,\lambda_5$	-9	-9	-9	-9
$g_1^2\,\lambda_5$	-9	-9	-3	-9
$\lambda_5\,T_\nu^{(1)}$	0	0	0	0
$\lambda_5\,T_\nu^{(2)}$	0	0	0	0
$\lambda_5\,\lambda_{HS_2}^{(A)}$	2	2	2	2
$\lambda_5\,T$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(A)}$	-2	-2	-2	-2
$\lambda_5\,\lambda_{12}^{(A)}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$
$\lambda_5\,T_\nu$	1	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$\lambda_5\,\lambda_{12}^{(\times)}$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$\lambda_5\,\lambda_1$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{HS_2}^{(1)}$	4	4	4	4
$\lambda_5\,\lambda_{12}^{(1)}$	2	2	2	2
$g_2^2\,\lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\,\lambda_5$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{9}{2}$
$\lambda_5\,T_\nu^{(1)}$	1	1	0	0
$\lambda_5\,T_\nu^{(2)}$	2	2	0	0
$\lambda_5\,\lambda_{HS_2}^{(A)}$	0	0	2	2
$\lambda_5\,T$	2	2	2	2
$\lambda_5\,\lambda_{HS_1}^{(A)}$	-4	-4	-2	-2
$\lambda_5\,\lambda_{12}^{(A)}$	0	0	2	2
$\lambda_5\,T_\nu$	0	0	1	1
$\lambda_5\,\lambda_{12}^{(\times)}$	0	0	-2	-2

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	$\mathrm{T3-E-p2}\ (\alpha=2)$
$\lambda_5\,\lambda_1$	2
$\lambda_5\,\lambda^{(1)}_{HS_1}$	4
$\lambda_5\,\lambda^{(1)}_{HS_2}$	4
$\lambda_5\,\lambda^{(1)}_{12}$	2
$g_2^2\,\lambda_5$	$-\frac{33}{2}$
$g_1^2\,\lambda_5$	$-\frac{33}{2}$
$\lambda_5\,T^{(1)}_{\nu}$	0
$\lambda_5\,T^{(2)}_{\nu}$	0
$\lambda_5\,\lambda^{(A)}_{HS_2}$	2
$\lambda_5\,T$	2
$\lambda_5\,\lambda^{(A)}_{HS_1}$	-2
$\lambda_5\,\lambda^{(A)}_{12}$	2
$\lambda_5\,T_{\nu}$	1
$\lambda_5\,\lambda^{(\times)}_{12}$	-2

Additional scalar-coupling beta functions

$\beta_{\lambda^{(A)}_{HS_1}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-m3}\ (\alpha=$
$g_1^2 g_2^2$	0	0	0	-18

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	$\mathrm{T3-B-p1}\ (\alpha=1)$	$\mathrm{T3-C-m3}\ (\alpha=-3)$	$\mathrm{T3-C-m1}\ (\alpha=-1)$	$\mathrm{T3-C-p1}\ (\alpha=$
$g_1^2 g_2^2$	6	-18	-6	6

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	$\mathrm{T3-D-p0}\ (\alpha=0)$	$\mathrm{T3-D-p2}\ (\alpha=2)$	$\mathrm{T3-E-m4}\ (\alpha=-4)$	$\mathrm{T3-E-m2}\ (\alpha=-$
$g_1^2 g_2^2$	0	12	-24	-12

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	$\mathrm{T3-E-p2}\ (\alpha=2)$
$g_1^2 g_2^2$	12

$\beta_{\lambda^{(A)}_{HS_2}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	$\mathrm{T3-A-m4}\ (\alpha=-4)$	$\mathrm{T3-A-m2}\ (\alpha=-2)$	$\mathrm{T3-A-p0}\ (\alpha=0)$	$\mathrm{T3-B-m3}\ (\alpha=$
$g_1^2 g_2^2$	-12	0	12	-6

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
$g_1^2 g_2^2$	18	−6	6	18

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$g_1^2 g_2^2$	0	0	−12	0

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$g_1^2 g_2^2$	24

$\beta_{\lambda_{12}^{(A)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B − m3 ($\alpha =$
$g_1^2 g_2^2$	0	0	0	18

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha =$
$g_1^2 g_2^2$	18	18	−6	18

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	T3 − D − p0 ($\alpha = 0$)	T3 − D − p2 ($\alpha = 2$)	T3 − E − m4 ($\alpha = -4$)	T3 − E − m2 ($\alpha = -$
$g_1^2 g_2^2$	0	0	48	0

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$g_1^2 g_2^2$	48

$\beta_{\lambda_{12}^{(\times)}}$

Run block 1: T3-A-m4–T3-B-m1

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B − m3 ($\alpha =$
g_2^4	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0

Run block 2: T3-B-p1–T3-D-m2

$\mathcal{S}$	$\mathrm{T3-B-p1}\ (\alpha=1)$	$\mathrm{T3-C-m3}\ (\alpha=-3)$	$\mathrm{T3-C-m1}\ (\alpha=-1)$	$\mathrm{T3-C-p1}\ (\alpha=$
g_2^4	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0
$\lambda_{12}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	0	0
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	0	0

Run block 3: T3-D-p0–T3-E-p0

$\mathcal{S}$	$\mathrm{T3-D-p0}\ (\alpha=0)$	$\mathrm{T3-D-p2}\ (\alpha=2)$	$\mathrm{T3-E-m4}\ (\alpha=-4)$	$\mathrm{T3-E-m2}\ (\alpha=-$
g_2^4	0	0	6	6
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	2	2
$\lambda_{12}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	8	8
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	10	10
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	-2	-2
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	-2	-2

Run block 4: T3-E-p2–T3-E-p2

$\mathcal{S}$	$\mathrm{T3-E-p2}\ (\alpha=2)$
g_2^4	6
$\left(\lambda_{12}^{(A)}\right)^2$	2
$\lambda_{12}^{(1)}\,\lambda_{12}^{(\times)}$	8
$\left(\lambda_{12}^{(\times)}\right)^2$	10
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	-2
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	-2

First threshold

The first EFT GroupFactors report below is selected from the actual threshold plan stored in the run. No assumption is made here about whether F, S_1, S_2, or a degenerate set is integrated out first.